

Supercaliprobabilisticexpialidocious

Understanding Household Electricity Demand Using Probability Theory

Abstract

This study investigates household electricity consumption using statistical inference and probability theory. The goal is to identify the best-fitting distribution, analyze temporal variations, and assess predictive relationships among electrical variables. Methods include goodness-of-fit tests, variance analysis, regression modeling, and categorical association tests. Results show non-normal characteristics, with Weibull providing the most suitable fit. Day type and season significantly affect demand, and Global Intensity emerges as the dominant predictor of active power. These findings support smart-grid optimization, demand-response strategies, and improved forecasting for energy management.

Objectives

Assess whether realistic synthetic hourly electricity-consumption data can be generated in a way that preserves the statistical characteristics of the observed real-world consumption.

- why
- Enables utilities and grid operators to improve 24-48 hour load forecasts, optimize generation mix, and reduce risk of overloads and blackouts.
 - Informs policy makers for targeted energy-conservation and renewable-integration policies based on household-level behavior. strategies.
 - Provides researchers with validated distributional models, synthetic datasets, and methods for anomaly detection, probabilistic forecasting, and extreme-event risk assessment.

- Questions
- Which probability distribution (Normal, Lognormal, Gamma, Weibull) best characterizes household electricity consumption?
 - Is consumption significantly different on weekends versus weekdays
 - Do seasonal variations significantly affect consumption patterns?
 - Can we accurately predict consumption from electrical variables?
 - Are categorical variables (hour, day type, season) independent of consumption levels?

Methodology

- Q1: Best-fitting distribution (Normal, Lognormal, Gamma, Weibull)?
Shapiro–Wilk: Checks normality for parametric assumptions.
Anderson–Darling: Evaluates fit with emphasis on tails.
Kolmogorov–Smirnov: Compares empirical vs theoretical CDF for candidate distributions.
Chi-Square GoF: Non-parametric check for binned data.
Why: Confirms distributional shape for accurate modeling.
- Q2: Is weekend consumption different from weekday?
Welch’s t-Test: Compares two means without assuming equal variances.
Why: Robust for large samples and variance differences.

- Q3: Do seasonal variations affect consumption?
One-Way ANOVA: Tests overall seasonal mean differences.
Tukey HSD: Identifies which seasons differ significantly.
Why: Detects and localizes seasonal effects.

- Q4: Can electrical variables predict consumption?
Simple Linear Regression: Tests intensity as main predictor.
Multiple Regression: Adds voltage and reactive power for incremental value.
Why: Quantifies predictive strength and model parsimony.

- Q5: Are categorical factors (hour, day type, season) independent of consumption?
Chi-Square Independence: Tests association between categories and consumption levels.
Why: Reveals temporal and behavioral influence on demand.

$$W = \frac{[\sum_{i=1}^n a_i x_{(i)}]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Shapiro–Wilk

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i - 1) [\ln F(x_i) + \ln (1 - F(x_{n+1-i}))]$$

Anderson–Darling

$$D = \max_x |F_n(x) - F(x)|$$

Kolmogorov–Smirnov

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Chi-Square Goodness-of-Fit Test (Weibull)

$$W = \frac{(N-k) \sum_{i=1}^k n_i (Z_i - Z_{n-i})^2}{(k-1) \sum_{i=1}^k \sum_{j=1}^{n_i} (Z_{ij} - Z_i)^2}$$

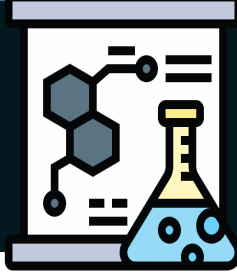
Tukey HSD

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Welch’s t-Test

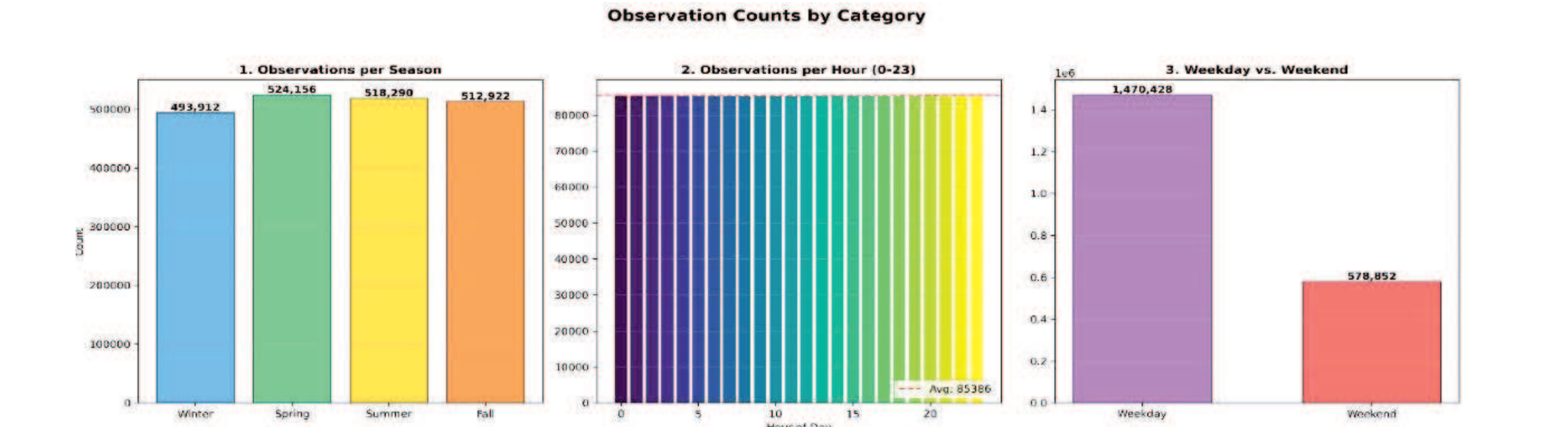
$$HSD = q \sqrt{\frac{MSE}{N}}$$

Tukey HSD

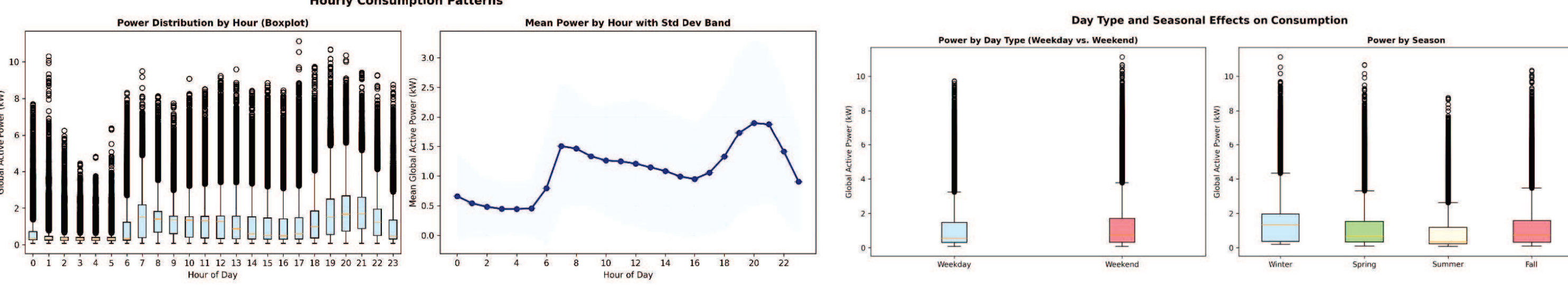


Results & Analysis

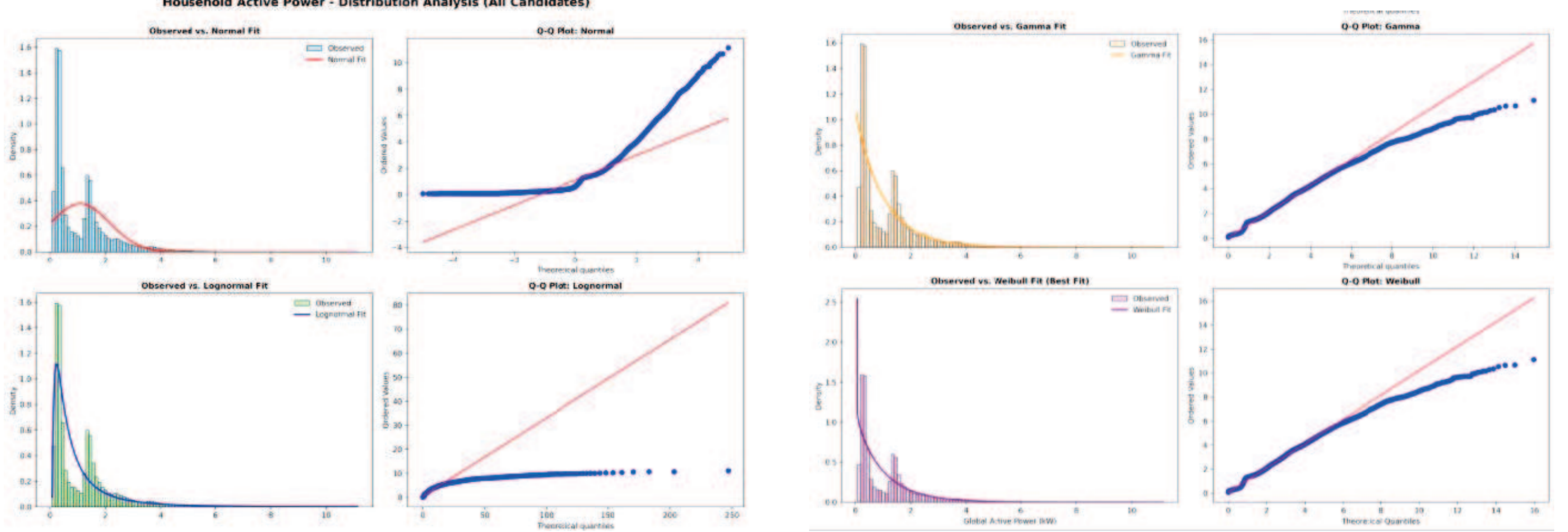
Balanced Dataset: Seasonal, hourly, and day-type distributions are nearly uniform, ensuring no major sampling bias and a solid foundation for temporal analysis.



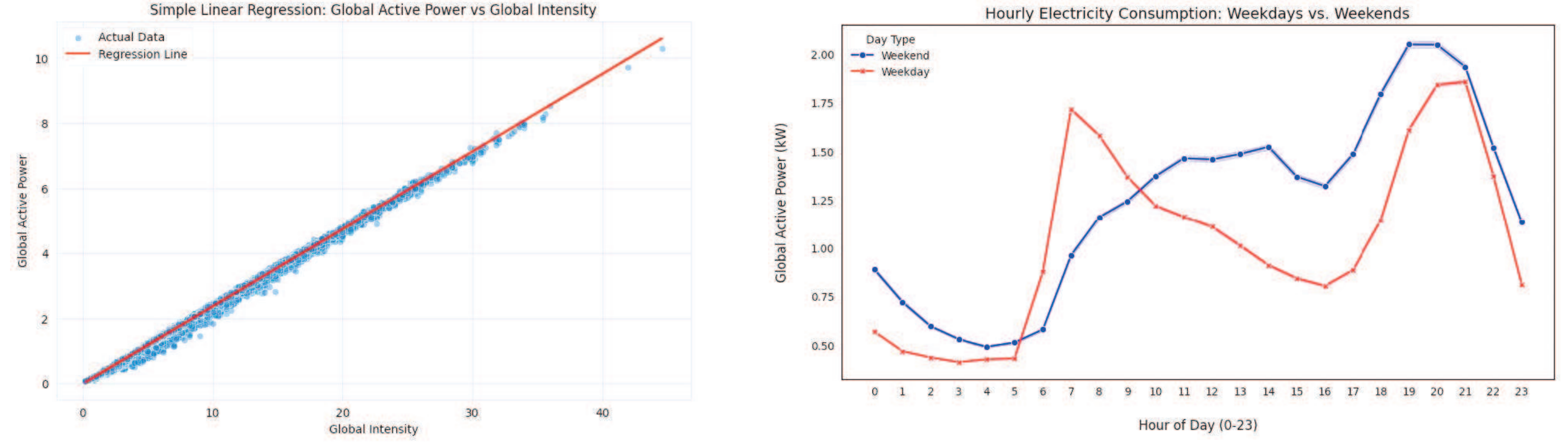
Daily Cycle: Consumption follows a clear pattern—low in early morning, rising through the day, peaking in the evening, with greater variability during peak hours.



Distribution Fit: Among Normal, Lognormal, Gamma, and Weibull, the Weibull distribution provides the best fit, closely matching density and Q–Q plots.



Predictive Relationship: Global Intensity shows a very strong correlation with Global Active Power, confirming it as the dominant predictor.



Conclusion and Future work

Household electricity consumption is best modeled by a Weibull distribution and exhibits clear temporal patterns by day type and season. A simple linear relationship between current intensity and active power enables accurate short-term forecasting. These insights support targeted demand-response programs and probabilistic planning for grid stability.

For Future Work

- Expand Scope: Analyze multiple households and appliance-level loads via sub-metering.
- Add Context: Integrate weather and occupancy data for better prediction.
- Validate Assumptions: Use Levene’s test to check variance equality before ANOVA.
- Non-Parametric Checks: Hourly: Kruskal–Wallis to confirm ANOVA results.
- Weekday vs Weekend: Mann–Whitney U to validate Welch’s t-test.
- Seasonal: Kruskal–Wallis with post-hoc Mann–Whitney (Bonferroni correction) for robust seasonal comparisons.

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